

CHE 201 Spring 2020 Final Exam

Roxanna Mendoza

TOTAL POINTS

85 / 100

QUESTION 1

1 15 pts

1.1 a 10 / 12

- **0 pts** Correct
- **2 pts** first is incorrect
- **2 pts** second is incorrect
- **2 pts** third is incorrect
- **2 pts** fourth is incorrect
- ✓ - **2 pts** fifth is incorrect
- **2 pts** sixth is incorrect

1.2 b 3 / 3

- ✓ - **0 pts** Correct
- **3 pts** Not done or wrong
- **2 pts** not explained

QUESTION 2

2 10 pts

2.1 a 5 / 5

- ✓ - **0 pts** Correct
- **1 pts** calc error
- **3 pts** h₂s is wrong
- **2 pts** formula is wrong
- **2 pts** explanation is not clear
- **5 pts** not done

2.2 b 5 / 5

- ✓ - **0 pts** Correct
- **2 pts** pr(T₁) is wrong
- **1 pts** calculation error
- **5 pts** not done or wrong
- **3 pts** formula is wrong

QUESTION 3

3 10 pts

3.1 a 2 / 2

- ✓ - **0 pts** Correct
- **2 pts** Not done or wrong

3.2 b 2 / 2

- ✓ - **0 pts** Correct
- **2 pts** wrong or not done
- **1 pts** one is wrong

3.3 c 2 / 6

- **0 pts** Correct
- **6 pts** not done or wrong
- ✓ - **2 pts** u value is wrong
- ✓ - **2 pts** s value is wrong

QUESTION 4

4 23 / 32

- **0 pts** Correct
- **2 pts** sketch is not drawn or incorrect
- **3 pts** engineering model is not listed
- **4 pts** simplified energy balance is not done
- **4 pts** work is not calculated or wrong
- **4 pts** heat is not calculated or wrong
- **5 pts** entropy change is not calculated
- **5 pts** sigma is not calculated
- ✓ - **2 pts** pv diagram is wrong
- ✓ - **3 pts** Ts diagram is wrong
- ✓ - **2 pts** v₂ is wrong
- **2 pts** u₂ is wrong
- ✓ - **2 pts** calculation error
- **2 pts** s₂ is wrong
- **3 pts** did not multiplied with mass
- **5 pts** v₁, u₁, s₁ are wrong
- **2 pts** T in sigma calc is wrong

QUESTION 5

5 5 33 / 33

✓ - 0 pts Correct

- 3 pts sketch is not drawn
- 3 pts engineering model is not listed
- 9 pts mass flow rate of inlet not calculated or

wrong

- 9 pts mass flow rate of outlet is not calculated
- 9 pts entropy production is not calculated
- 2 pts s_3 is wrong
- 2 pts h_3 is wrong
- 2 pts h_2 is wrong
- 2 pts s_2 is wrong

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#1

FALSE

FALSE

TRUE

TRUE

TRUE

TRUE

Part B:

→ less than zero

$$\Delta S_{AB} = 0 \quad \text{because} \quad Q_{AB} = 0 \quad \& \quad \sigma = 0$$

$\uparrow$
 adiabatic

$\uparrow$
 reversible

$$\Delta S_{\text{total}} = 0 \text{ because it is a cycle}$$

$$\frac{\delta Q}{T} = -\sigma \quad \sigma \text{ irreversible means positive value}$$

So Q should be less than zero.

1.1 a 10 / 12

- 0 pts Correct
- 2 pts first is incorrect
- 2 pts second is incorrect
- 2 pts third is incorrect
- 2 pts fourth is incorrect
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- 2 pts sixth is incorrect

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So Q should be less than zero.

1.2 b 3 / 3

✓ - **0 pts** Correct

- **3 pts** Not done or wrong

- **2 pts** not explained

#2.

$$P_1 = 10 \text{ bar} \quad T_1 = 300^\circ\text{C}$$

$$P_2 = 5 \text{ bar}$$

$$W = 150 \text{ kJ/kg}$$

$$\frac{300 - 280}{320 - 280} = \frac{h_1 - 3008.2}{3093.9 - 3008.2}$$

$$0.5 = \frac{h_1 - 3008.2}{85.7}$$

$$42.85 = h_1 - 3008.2$$

$$h_1 = 3051.05 \text{ kJ/kg}$$

$$\eta_t = \frac{\dot{W}_{cv}/\dot{m}}{(\dot{W}_{cv}/\dot{m})_s} = \frac{150 \text{ kJ/kg}}{h_1 - h_{2s}}$$

$$S_{2s} = S_1$$

$$\frac{300 - 280}{320 - 280} = \frac{S_1 - 7.0465}{7.1962 - 7.0465}$$

$$S_1 = 7.12135 \text{ kJ/kg}$$

$$S_{2s} = 7.12135 \text{ kJ/kg}$$

$$\frac{T - 200}{240 - 200} = \frac{7.12135 - 7.0592}{7.2307 - 7.0592} \quad T = 214.5^\circ\text{C}$$

$$\frac{214.5 - 200}{240 - 200} = \frac{h_{2s} - 2055.4}{2939.9 - 2055.4}$$

$$h_{2s} = 2880.02 \text{ kJ/kg}$$

$$\eta_T = \frac{150 \text{ kJ/kg}}{3051.05 \text{ kJ/kg} - 2880.02 \text{ kJ/kg}} = 0.91$$

Part B: $T_1 = 650 \text{ K}$, $P_1 = 10 \text{ bar}$
 $S_2 = S_1$, $P_2 = 2 \text{ bar}$

$$S(T_2, P_2) - S(T_1, P_1) = C_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

$$Pr(T_1) = 21.906$$

$$Pr(T_2) = Pr(T_1) \left(\frac{P_2}{P_1} \right) = 21.906 \left(\frac{2 \text{ bar}}{10 \text{ bar}} \right) = 4.372$$

$$\frac{T - 410}{420 - 410} = \frac{4.372 - 4.153}{4.522 - 4.153}$$

$$T_2 = 415.93 \text{ K}$$

2.1a 5 / 5

✓ - 0 pts Correct

- 1 pts calc error
- 3 pts h2s is wrong
- 2 pts formula is wrong
- 2 pts explanation is not clear
- 5 pts not done

$$\frac{214.5 - 200}{240 - 200} = \frac{h_{2s} - 2055.4}{2939.9 - 2055.4}$$

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$$T_2 = 415.93 \text{ K}$$

2.2 b 5 / 5

✓ - 0 pts Correct

- 2 pts $\text{pr}(T1)$ is wrong
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- 3 pts formula is wrong

#3.

Part A: $g(T, P):$

$$\left(\frac{\partial v}{\partial T} \right)_P = - \left(\frac{\partial s}{\partial P} \right)_T$$

Part B:

$$dh = Tds + vdp$$

$$\frac{\partial h}{\partial s \partial P} = \left(\frac{\partial T}{\partial P} \right)_s$$

$$\frac{\partial h}{\partial P \partial s} = \left(\frac{\partial v}{\partial s} \right)_P$$

Part C:

$$m = 3 \text{ kg}$$

$$P = 10 \text{ bar}$$

$$T = 180^\circ\text{C} \Rightarrow 423 \text{ K}$$

$$\psi = u - Ts$$

Assumption sat liq.

$$u = 761.7 \text{ kJ/kg}$$

$$s = 2.1307 \text{ kJ/kg}\cdot\text{K}$$

$$\psi = (761.7) - (423 \text{ K}) \cdot 2.1307 =$$

$$\psi = 3 \times -142.97 = -428.91 \text{ kJ}$$

3.1 a 2 / 2

✓ - 0 pts Correct

- 2 pts Not done or wrong

#3.

Part A: $g(T, P):$

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3.2 b 2 / 2

✓ - **0 pts** Correct

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3.3 C 2 / 6

- 0 pts Correct
- 6 pts not done or wrong
- ✓ - 2 pts u value is wrong
- ✓ - 2 pts s value is wrong

#4.



$$m = 3 \text{ kg}$$

$$T_1 = 160^\circ\text{C}$$

$$P_1 = 3 \text{ bar}$$

$$x = 0.6$$

Engineering model

① closed system

② $\Delta KE = 0$ $\Delta PE = 0$

③ Isobaric (constant pressure)

④ work is based on change in volume

⑤ $Q \neq 0$

Energy Balance: $\Delta E = \Delta U + \Delta PE + \Delta KE$

$$Q - W = \Delta U$$

Entropy Balance:

$$\Delta S = \int \frac{\delta Q}{T} + \sigma$$

$$W = \int P dV = P (V_2 - V_1) = P m (v_2 - v_1)$$

$$v_1 = 0.651 \text{ m}^3/\text{kg}$$

$$v_2 = (1-x)v_f + x v_g \Rightarrow$$

$$\downarrow \quad \quad \quad \searrow$$

$$1.0732 \times 10^{-3} \quad 0.6058$$

$$v_2 = (1-0.6)(1.0732 \times 10^{-3} \text{ m}^3/\text{kg}) + 0.6(0.6058 \text{ m}^3/\text{kg})$$

$$v_2 = 429.64 \text{ m}^3/\text{kg}$$

$$W = 3 \text{ bar} \times \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} \times 3 \text{ kg} \times (429.64 - 0.651)$$

$$W = 386090.1 \text{ kJ}$$

$$\Delta U = Q - W$$

$$m(u_2 - u_1) = Q - W$$

$$u_1 = 2587.1 \text{ kJ/kg}$$

$$u_2 = (1-x)u_f + xu_g \quad @ \quad p = 3 \text{ bar}$$

$$u_2 = (1-0.6)561.15 + (0.6)(2543.6)$$

$$u_2 = 1750.62 \text{ kJ/kg}$$

$$3 \cdot (1750.62 - 2587.1) = Q - 386090.1$$

$$Q = 383580.66 \text{ kJ}$$

$$s_1 = 7.1276 \text{ kJ/kg} \cdot \text{K}$$

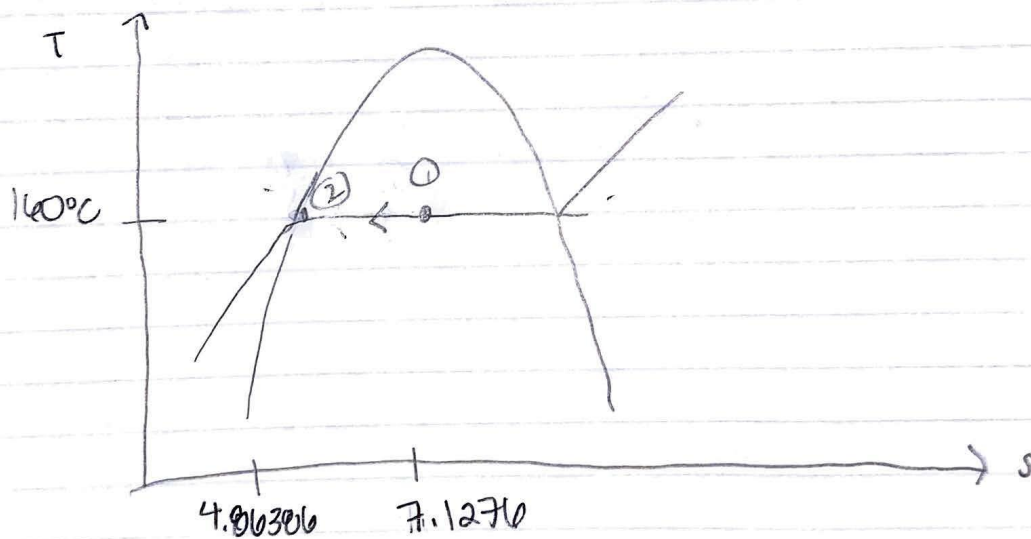
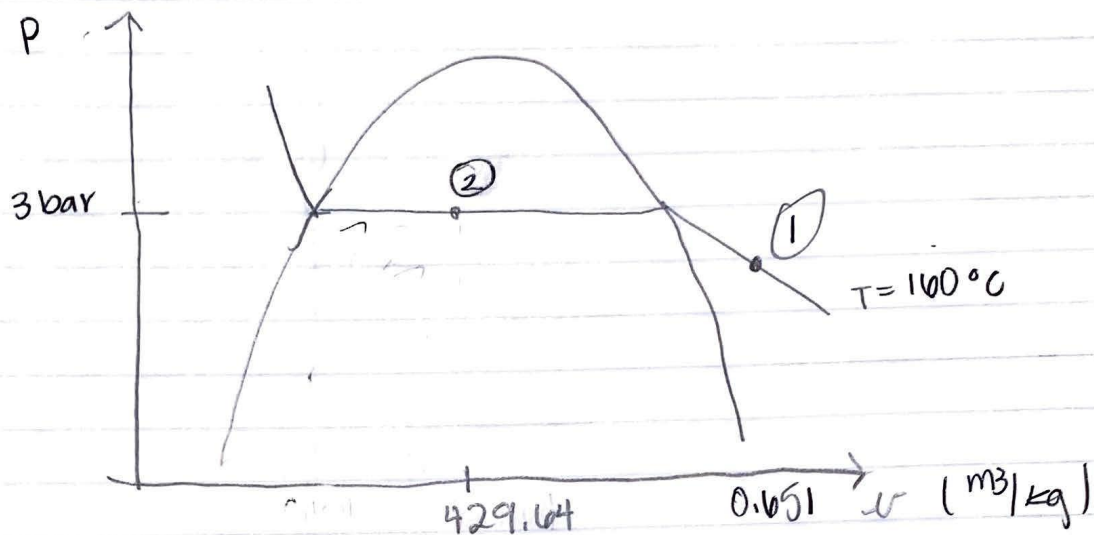
$$s_2 = (1-0.6)(1.6718) + (0.6)(6.9919) = 4.86386 \text{ kJ/kg} \cdot \text{K}$$

$$\Delta S = m(s_2 - s_1) = 3 \text{ kg} \cdot (4.86386 - 7.1276)$$

$$\Delta S = -6.79122$$

$$\Delta S = \frac{Q}{T_0} + \sigma \quad \Rightarrow \quad -6.79122 = \frac{383580.66}{298 \text{ K}} + \sigma$$

$$\sigma = -1293.97 \rightarrow \text{impossible}$$



4 4 23 / 32

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- 5 pts entropy change is not calculated
- 5 pts sigma is not calculated
- ✓ - 2 pts pv diagram is wrong
- ✓ - 3 pts Ts diagram is wrong
- ✓ - 2 pts v2 is wrong
 - 2 pts u2 is wrong
- ✓ - 2 pts calculation error
 - 2 pts s2 is wrong
 - 3 pts did not multiplied with mass
 - 5 pts v1,u1,s1 are wrong
 - 2 pts T in sigma calc is wrong

①

②

Engineering Model

① Open system

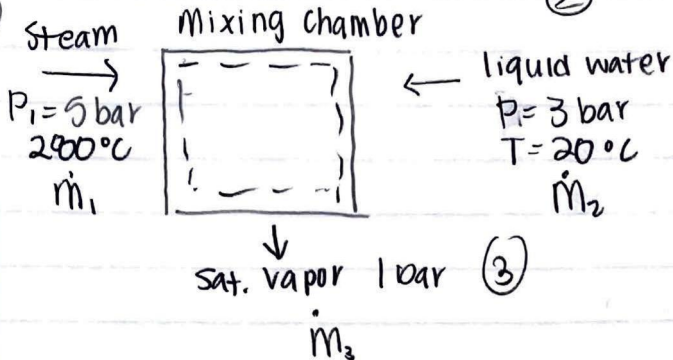
② Steady state

③ $\Delta KE = \Delta PE = 0$

④ Two inlets; one outlet

⑤ $Q_{cv} = 0$ adiabatic

⑥ $W_{cv} = 0$



Energy Balance:

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_3 h_3 = 0 \leftarrow \text{steady state}$$

Mass Balance:

$$\dot{m}_1 + \dot{m}_2 - \dot{m}_3 = 0 \leftarrow \text{steady state}$$

Entropy Balance:

$$\dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{Q}_{cv} = 0 \leftarrow \text{steady state}$$

$$h_1 = 3022.9 \text{ kJ/kg} \leftarrow \text{From Table}$$

$$h_2 \approx h_f(20^\circ\text{C}) = 83.96 \text{ kJ/kg}$$

$$h_3 = 2675.5 \text{ kJ/kg}$$

$$\dot{m}_1 = \left(20,000 \frac{\text{kg}}{\text{min}} \right) \left(\frac{1 \text{ min}}{60 \text{ sec}} \right) = 333.3 \text{ kg/sec}$$

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 - (\dot{m}_1 + \dot{m}_2) h_3 = 0$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_1 h_3 - \dot{m}_2 h_3 = 0$$

$$\dot{m}_1 (h_1 - h_3) + \dot{m}_2 (h_2 - h_3) = 0$$

$$\dot{m}_2 = \dot{m}_1 \left(\frac{h_3 - h_1}{h_2 - h_3} \right)$$

$$\dot{m}_2 = (333.3) \left(\frac{2675.5 - 3022.9}{83.96 - 2675.5} \right)$$

$$\dot{m}_2 = 44.68 \text{ kg/s}$$

liquid water
← entering chamber

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\dot{m}_3 = (333.3) + (44.68)$$

$$\dot{m}_3 = 378.01 \text{ kg/s}$$

← saturated vapor
exiting chamber

$$s_1 = 7.3865 \text{ kJ/kg} \cdot \text{K}$$

$$s_2 = 0.2966 \text{ kJ/kg} \cdot \text{K}$$

$$s_3 = 7.3594 \text{ kJ/kg} \cdot \text{K}$$

$$-\dot{Q}_{cv} = \dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3$$

$$-\dot{Q}_{cv} = (333.33)(7.3865) + (44.68)(0.2966) - (378.01)(7.3594)$$

$$-\dot{Q}_{cv} = -306.53 \text{ kJ/s} \Rightarrow \dot{Q}_{cv} = 306.53 \text{ kJ/s}$$

5 5 33 / 33

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- 9 pts mass flow rate of outlet is not calculated
- 9 pts entropy production is not calculated
- 2 pts s_3 is wrong
- 2 pts h_3 is wrong
- 2 pts h_2 is wrong
- 2 pts s_2 is wrong